

Laboratory Assignment 7: Logic Design with Combinational Building Blocks

ECE 0201: Digital Circuits and Systems (50 Points)

Name

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Submission Checklist:

- ☐ Write within boxes, do not move boxes
- ☐ Write your full name in the box above
- ☐ Save this file as a PDF before uploading, keep the number of pages (**10**) unchanged
- ☐ Note “TO BE CONTINUED” in the answer box if you used the extra pages (9-10)

Part I: 4-Bit BCD to 7-Segment Display (10 points)

[(A) Show your derivation of logic functions for 7-segment display segments.] (3 points)

	A B C D	a	b	c	d	e	f	g
0	0 0 0 0	1	1	1	1	1	1	0
1	0 0 0 1	0	1	1	0	0	0	0
2	0 0 1 0	1	1	0	1	1	0	1
3	0 0 1 1	1	1	1	1	0	0	1
4	0 1 0 0	0	1	1	0	0	1	1
5	0 1 0 1	1	0	1	1	0	1	1
6	0 1 1 0	1	0	1	1	1	1	1
7	0 1 1 1	1	1	1	0	0	0	0
8	1 0 0 0	1	1	1	1	1	1	1
9	1 0 0 1	1	1	1	1	0	1	1

$$a = A + C + BD + B'D'$$

$$b = B' + C'D' + CD$$

$$c = B + C' + D$$

$$d = A + B'D' + B'C + CD' + BC'D$$

$$e = B'D' + CD'$$

$$f = A + C'D' + BC' + BD'$$

$$g = A + BC' + B'C + CD'$$

[(B) Insert a screenshot of your VHDL code for Part I.] (3 points)

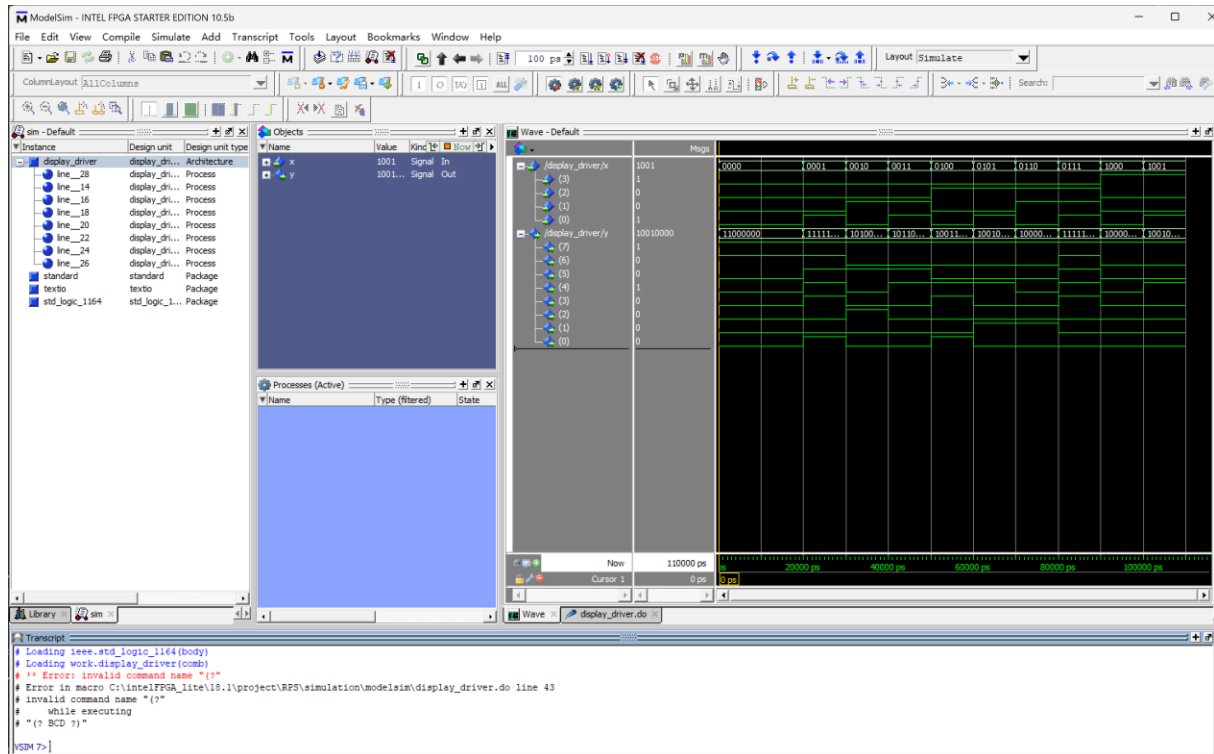
```
1  LIBRARY ieee;
2  USE ieee.std_logic_1164.all;
3
4  ENTITY display_driver IS
5  PORT(
6    x : IN  STD_LOGIC_VECTOR(3 DOWNTO 0);
7    y : OUT STD_LOGIC_VECTOR(7 DOWNTO 0)
8  );
9  END display_driver;
10
11 ARCHITECTURE comb OF display_driver IS
12 BEGIN
13
14   y(0) <= (x(3) AND x(2)) OR (x(3) AND x(1)) OR ((NOT x(3)) AND (NOT x(2)) AND (NOT x(1)) AND x(0)) OR (x(2) AND (NOT x(1)) AND (NOT x(0)));
15   y(1) <= (x(3) AND x(2)) OR (x(3) AND x(1)) OR (x(2) AND (x(1) XOR x(0)));
16   y(2) <= (x(3) AND x(2)) OR (x(3) AND x(1)) OR ((NOT x(2)) AND x(1) AND (NOT x(0)));
17   y(3) <= (x(3) AND x(2)) OR (x(3) AND x(1)) OR ((NOT x(3)) AND (NOT x(2)) AND (NOT x(1)) AND x(0)) OR (x(2) AND (NOT (x(1) XOR x(0))));
18   y(4) <= x(0) OR (x(2) AND (NOT x(1)) AND (NOT x(0))) OR (x(3) AND x(1) AND (NOT x(0)));
19   y(5) <= (x(3) AND x(2)) OR (x(3) AND x(1)) OR (x(1) AND x(0)) OR ((NOT x(3)) AND (NOT x(2)) AND (x(1) OR x(0)));
20   y(6) <= (x(3) AND x(2)) OR (x(3) AND x(1)) OR (x(2) AND x(1) AND x(0)) OR ((NOT x(3)) AND (NOT x(2)) AND (NOT x(1)));
21   y(7) <= '1';
22
23   comb;
24
25 END comb;
26
27
28
29
30
31
```

Messages

- 22036 Successfully launched Nativelink simulation (quartus_sh -t "c:/intelFPGA_lite/18.1/quartus/common/tcl/internal/nativelink/qnativesim.tcl" --rtl_sim "display_driver" "display_dri
- 22036 For messages from Nativelink execution see the Nativelink log file c:/intelFPGA_lite/18.1/project/display_driver/display_driver_nativelink_simulation.rpt

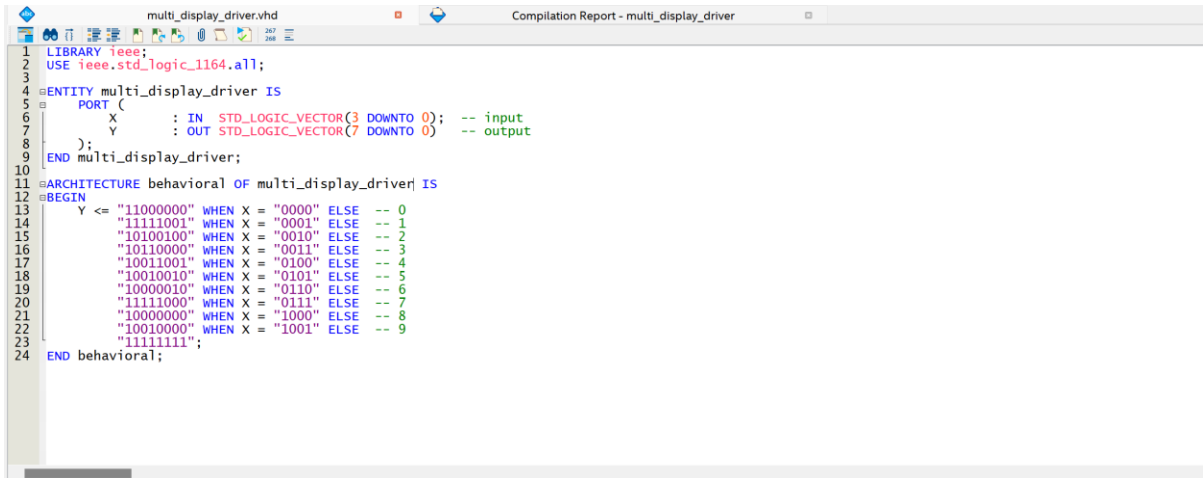
System (2) Processing (122)

[(B) Insert a screenshot of your ModelSim simulation for Part I.] (4 points)



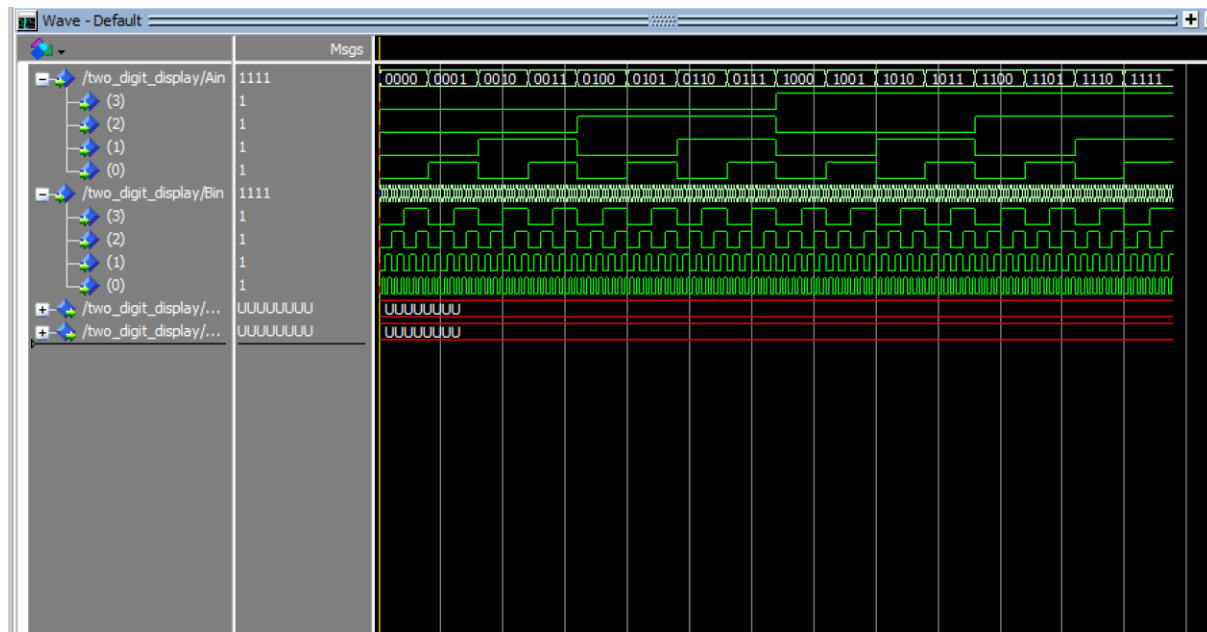
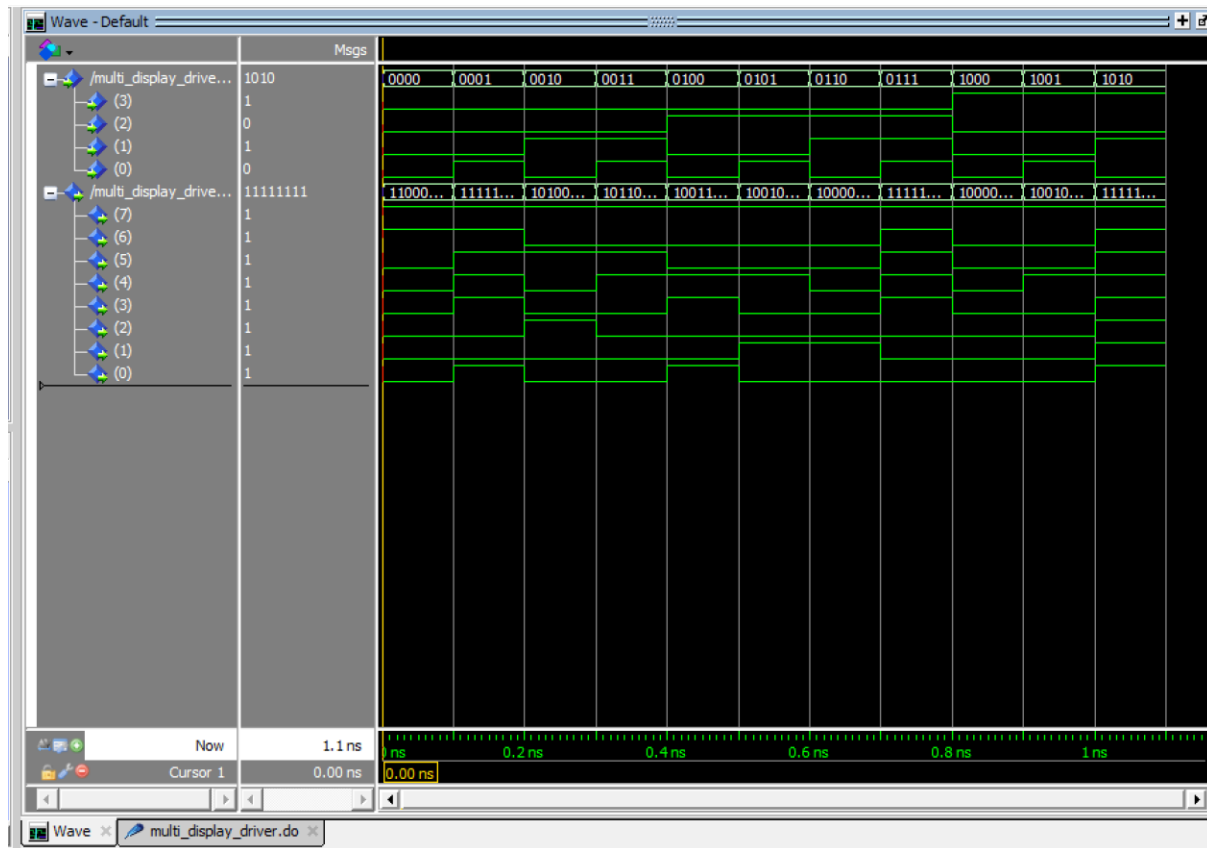
Part II: Multiple 7-segment Display Drivers (32 points)

[(A) Insert a screenshot of your VHDL code for Part II.] (2 points)



```
1  LIBRARY ieee;
2  USE ieee.std_logic_1164.all;
3
4  ENTITY multi_display_driver IS
5      PORT (
6          X      : IN  STD_LOGIC_VECTOR(3 DOWNTO 0); -- input
7          Y      : OUT STD_LOGIC_VECTOR(7 DOWNTO 0); -- output
8      );
9  END multi_display_driver;
10
11 ARCHITECTURE behavioral OF multi_display_driver IS
12 BEGIN
13     Y <= "11000000" WHEN X = "0000" ELSE -- 0
14         "11111001" WHEN X = "0001" ELSE -- 1
15         "10100100" WHEN X = "0010" ELSE -- 2
16         "10110000" WHEN X = "0011" ELSE -- 3
17         "10011001" WHEN X = "0100" ELSE -- 4
18         "10010010" WHEN X = "0101" ELSE -- 5
19         "10000010" WHEN X = "0110" ELSE -- 6
20         "11111000" WHEN X = "0111" ELSE -- 7
21         "10000000" WHEN X = "1000" ELSE -- 8
22         "10010000" WHEN X = "1001" ELSE -- 9
23         "11111111";
24 END behavioral;
```

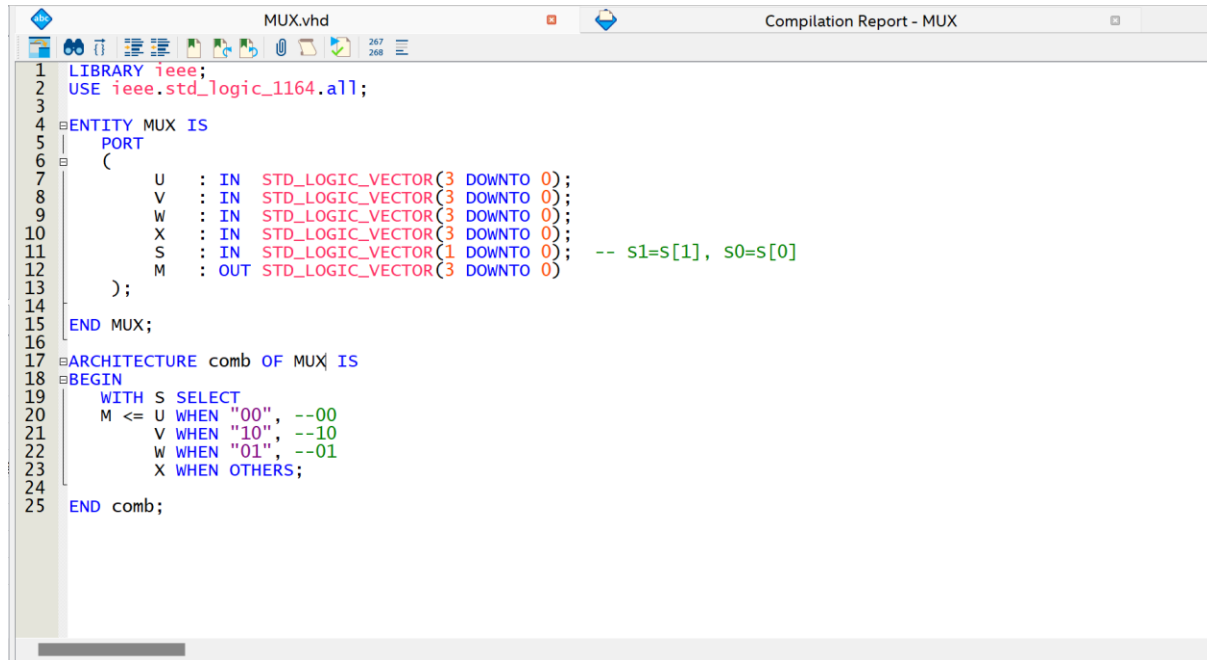
[(B) Insert a screenshot of your ModelSim simulation for Part II.] (2 points)



[(C) Demonstrate your circuit to the TA for Lab 7 check-off.] (28 points)

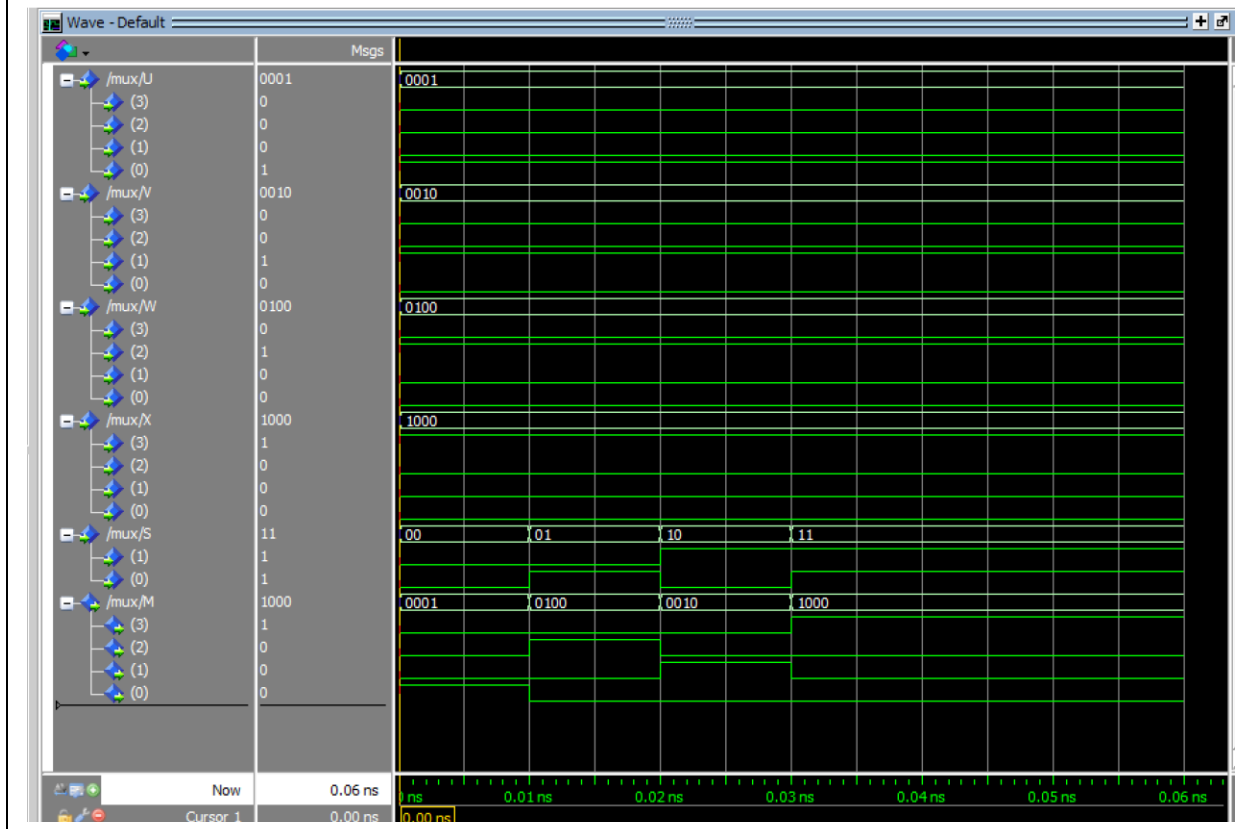
Part III: 4-bit 4:1 MUX (8 points)

[(A) Insert a screenshot of your VHDL code for Part III.] (3 points)



```
1  LIBRARY ieee;
2  USE ieee.std_logic_1164.all;
3
4  ENTITY MUX IS
5      PORT
6      (
7          U : IN  STD_LOGIC_VECTOR(3 DOWNTO 0);
8          V : IN  STD_LOGIC_VECTOR(3 DOWNTO 0);
9          W : IN  STD_LOGIC_VECTOR(3 DOWNTO 0);
10         X : IN  STD_LOGIC_VECTOR(3 DOWNTO 0);
11         S : IN  STD_LOGIC_VECTOR(1 DOWNTO 0); -- s1=s[1], s0=s[0]
12         M : OUT STD_LOGIC_VECTOR(3 DOWNTO 0);
13     );
14
15 END MUX;
16
17 ARCHITECTURE comb OF MUX IS
18 BEGIN
19     WITH S SELECT
20         M <= U WHEN "00", --00
21              V WHEN "10", --10
22              W WHEN "01", --01
23              X WHEN OTHERS;
24
25 END comb;
```

[(B) Insert an image of your ModelSim Simulation.] (4 points)



[(C) Insert the text of your do file.] (1 point)

```
Ln#
1 vlib work
2 vmap work work
3
4 vcom C:/intelFPGA_lite/18.1/project/MUX/MUX.vhd
5
6 vsim -voptargs="+acc" work.MUX
7
8 add wave *
9
10 force U b"0001"
11 force V b"0010"
12 force W b"0100"
13 force X b"1000"
14
15 # S = 00 ? M = U
16 force S b"00"
17 run 10
18
19 # S = 01 ? M = W
20 force S b"01"
21 run 10
22
23 # S = 10 ? M = V
24 force S b"10"
25 run 10
26
27 # S = 11 ? M = X
28 force S b"11"
29 run 10
30
31 run 20
```


[Leave the page unchanged if you are not going to use. Otherwise, specify the problem number you are working on here. Don't forget to notify "TO BE CONTINUED" in the original problem box]

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